Sequence Mapping and Visualisation

Infectious Disease 'Omics Course
April 2025

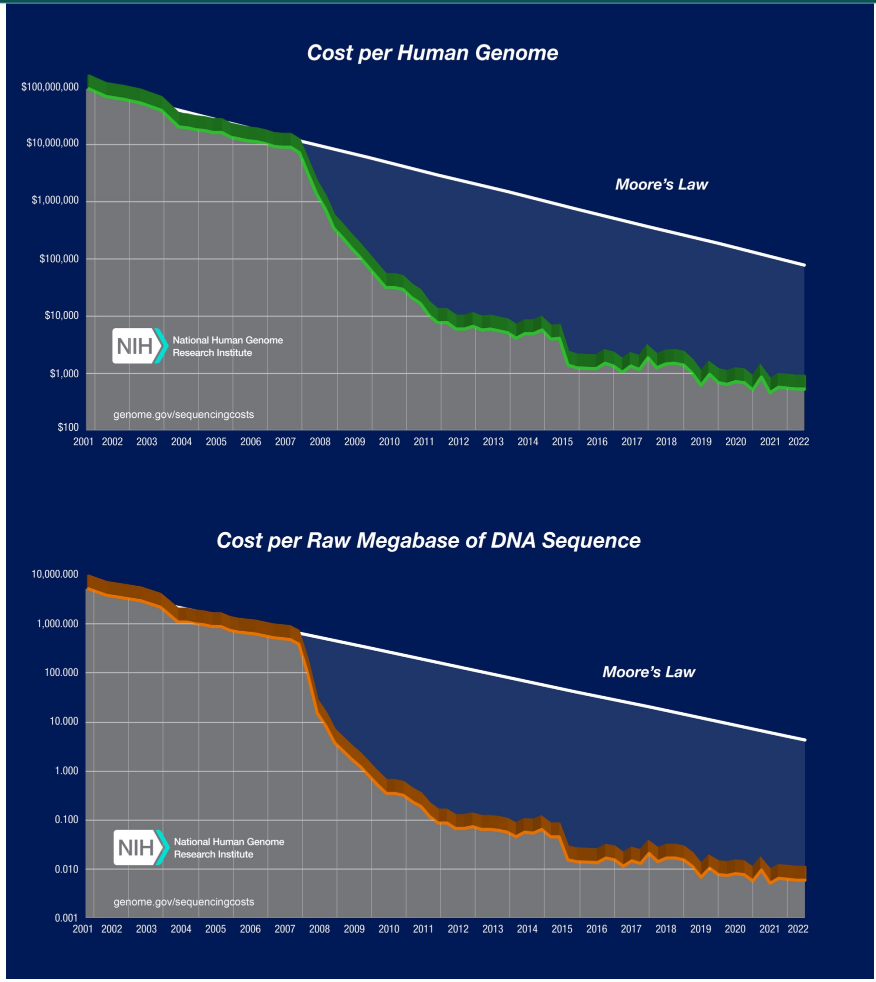
LONDON SCHOOL of HYGIENE & TROPICAL MEDICINE

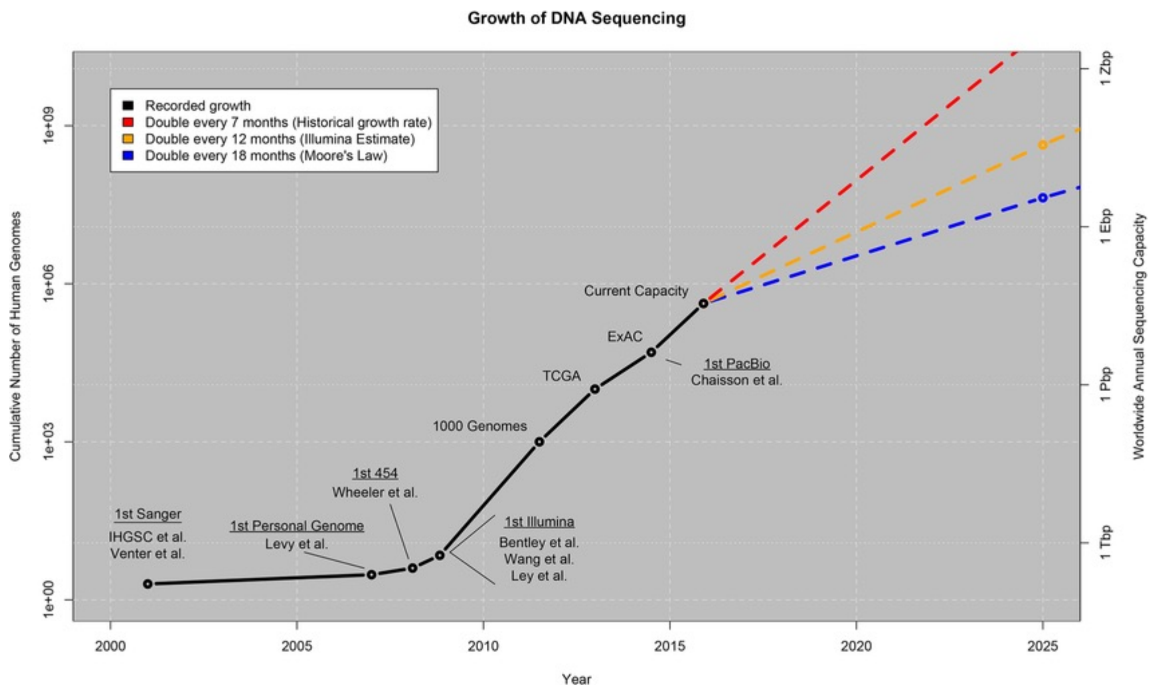
Outline

This lecture will cover:

- How we map sequence data to a reference genome
- How we can then assess the quality of the alignment
- Putting this into practice!

Sequencing data is more available than ever





Sources:
https://www.genome.gov/about-genomics/fact-sheets/DNA-Sequencing-Costs-Data
Stephens et al (2015) Big Data: Astronomical or Genomical? PLOS Biology.

'Next Generation' Sequencing

- Next generation sequencing is high throughput and much faster than older sequencing methods such as capillary sequencing
- The most common type of next generation sequencing is Illumina

Capillary or 'Sanger' Sequencing

1970s
Run time: months

→

'Next Generation Sequencing'

Early 2000s
Massive parallelization
Run time: hours

→

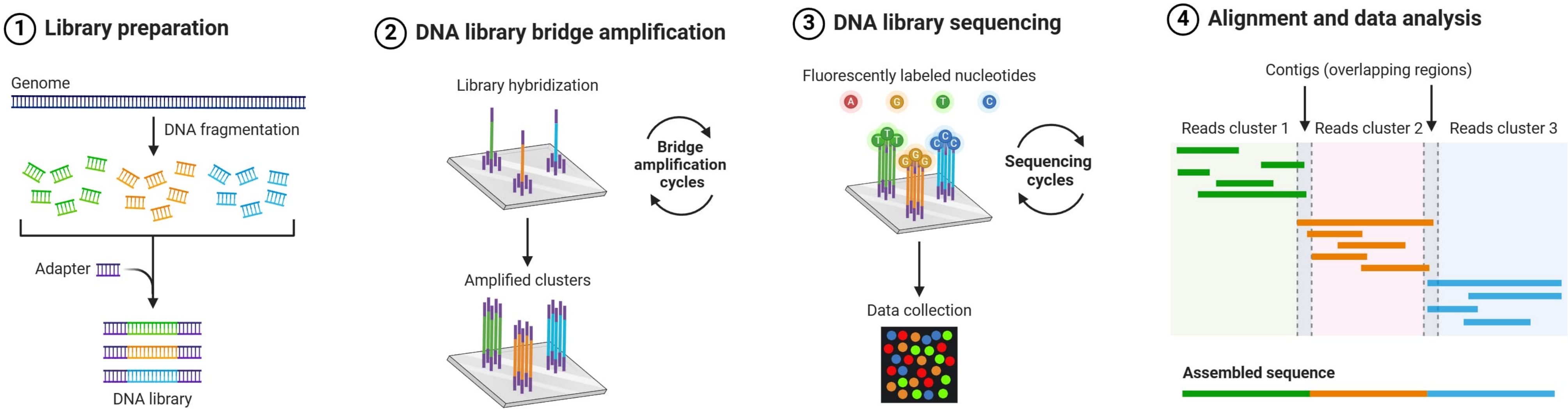
'Third Generation Sequencing'

2014 onwards
Run time: hours

Oxford Nanopore Technologies, PacBio

Illumina sequencing produces short read data

- The most commonly generated sequence data is Illumina Next-Generation data.



Also known as Cluster Generation

Source image: https://microbenotes.com/illumina-sequencing/

FASTQ files are produced containing the read data

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

- For each cluster that is sequenced, a sequence is written to the corresponding samples FASTQ file
- For a paired-end sequencing run, one R1 FASTQ file and one R2 FASTQ file are created for each cluster, as the DNA fragment is sequenced from both ends
- Each entry in a FASTQ file has 4 lines

Sequence identifier

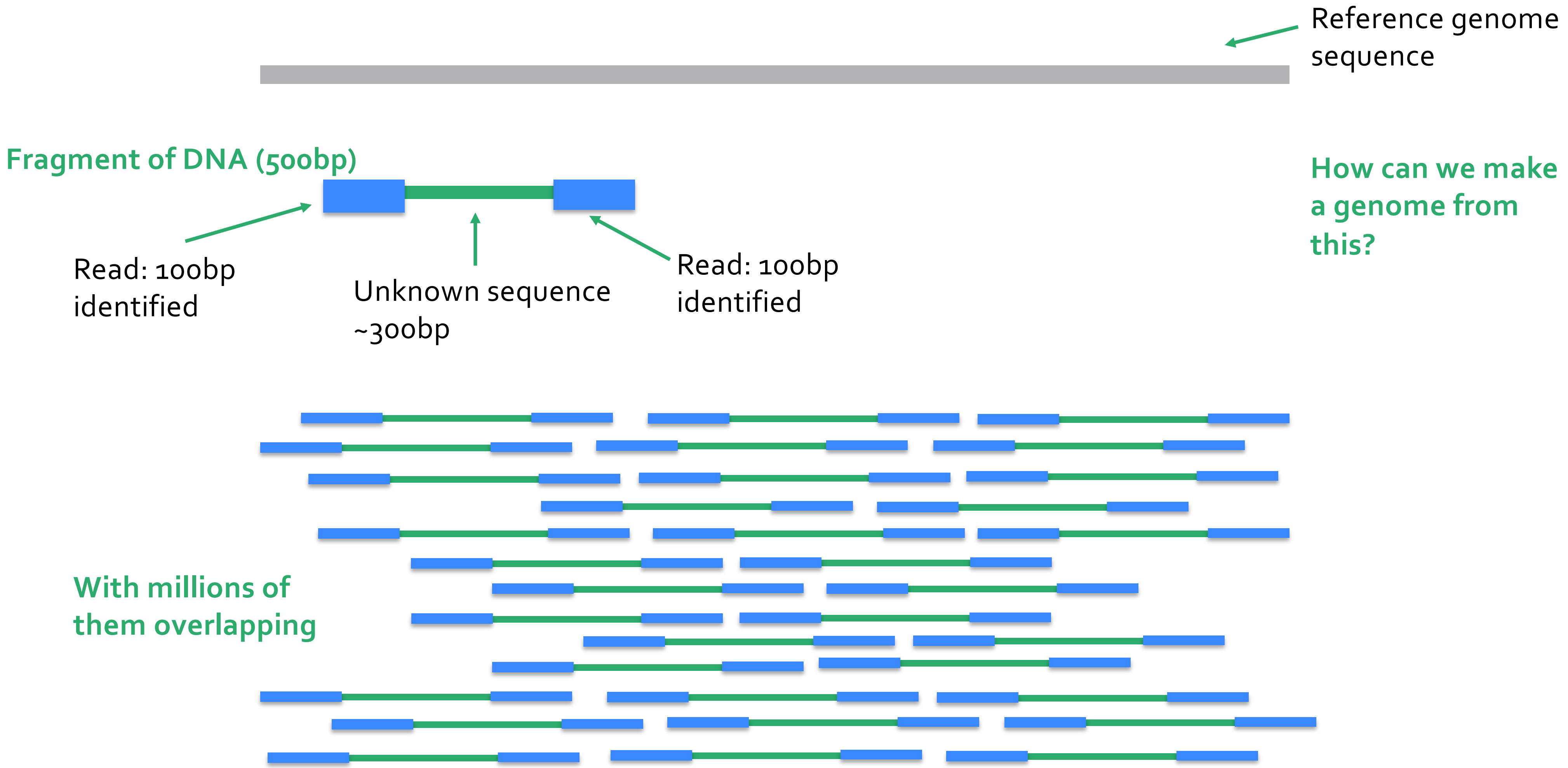
The sequence of bases (A, C, T, G)

Base call quality scores

A separator + sign

```
@HS4_5964:1:1106:17017:101018#12/1
CCTTAGGGTCGCCGTTAAGTTCGGAGACGACCGCGTTCCACACTGTGGTGAAGCCTGAACCGGGGTCATCGGTCA
•+
GDDGGE@E?B????BDEEEE?=DBBGDGGDD?BBBDGAGDB:=B=CE?9EDAGD@===:=292,/9=:=B=566;
@HS4_5964:1:2205:13272:35605#12/1
CCTTAGGGTCGCCGTTAAGTTCGGAGACGACCGCGTTCCACACTGTGGTGAAGCCTGAACCGGGGTCATCGGTCA
•+
IIIIHIHIIIIIIIIHIIIHIIFIFHIIIIHIIIIIIIIDIDBGGGGDG>DGBD@DGGGDFG>E?C2>D<8B(*,46
@HS4_5964:1:2108:7021:12911#12/1
ACCTTAGGGTCGCCGTTAAGTTCGGAGACGACCGCGTTCCACACTGTGGTGAAGCCTGAACCGGGGTCATCGGTC
•+
IIIIIHIIHIIIIIIHIHIIIIIIGIDIFIIIIIIIIHIIHIHIIEIDFIHIGFIIIIIG>GEE?2CCEFG8BD
@HS4_5964:1:1206:21270:179616#12/1
CAANCTTAGGGTCGCCGTTAAGTTCGGAGACGACCGCGTTCCACACTGTGGTGAAGCCTGAACCGGGGTCATCGG
•+
>>7%<??;?8DGGBGEGGDCDGG>GHHHHHGD@@@DGGGGHDHBHBGDECDDGD<EE??<?=?A:+744'=;947
@HS4_5964:1:1103:11932:160767#12/1
AAACGGCACTCGACAATCAAGCGAGGATGGCGGATTGACTAGCGGGCCCGACAACCTGGACCCGGGGGTTTCAAC
•+
GGGDGGGEGGGEGDG2BBB=BBD?DGGGGG4<=/<'/18-+550-('1)-+4-1.',,)6(.&11&2)(7.&',4
@HS4_5964:1:2206:3766:101157#12/1
CGTCGTCAACCTTAGGGTCGCCGTTAAGTTCGGAGACGACCGCGTTCCACACTGTGGTGAAGCCTGAACCGGGGT
•+
IIIIIIIIIIIIIIIIIHIIGIIIIIIIIIIGIBIFHIHIIIIIHIIIGIIGGIHHIDBDIIIIHHIDIGGID
@HS4_5964:1:2104:10206:46786#12/1
GGGTGTTTTCAACACGAGGATCACGAGCCGTTGCCGGTAGGTTGCCGCTGGGTTTTGTAGGGGAGGTCTACCAAT
•+
BC?CAFGEAAGGGGDDG@BGEFBFEGD:GGGGG:B7;;?3;3;31+:32/>+)'')'&&*4'*))&)'&&1')'7
@HS4_5964:1:1206:14745:64142#12/1
GCTGGGTCCGTCGTCAACCTTAGGGTCGCCGTTAAGTTCGGAGACGACCGCGTTCCACACTGTGGTGAAGCCTGA
•+
IIIIIIIIIIIIIIHIIIIIIIIIHIGIIIIIIHFIIGIEGIIIHHIIDIIIIIIIIHIIHHIIIIHIGFIIII
```

Illumina produces millions of read fragments



How can we make a genome from this?

Reads are aligned to a reference

Single reads in green

Paired end reads in blue

Reference Genome

LookSeq visualisation tool

Read alignment can be done with several tools

- Alignment to a reference (e.g. BWA, Bowtie)
- Multiple alignment format (e.g. BAM, SAM, CRAM)
- Visualising read alignment and variants in browsers (e.g. Tablet, Artemis, IGV)
- Algorithms for calling variants
  - Small variants (e.g. SNPs, indels) GATK/Freebayes → VCF
  - Larger variants (e.g. large indels) Pindel, Delly, Lumpy
- Reference free or *de novo* assembly (e.g. Velvet)

Quality Control is essential

- Screening for contamination
  - E.g. search a large sequence dataset against a panel of different databases (*eg. Kraken*)

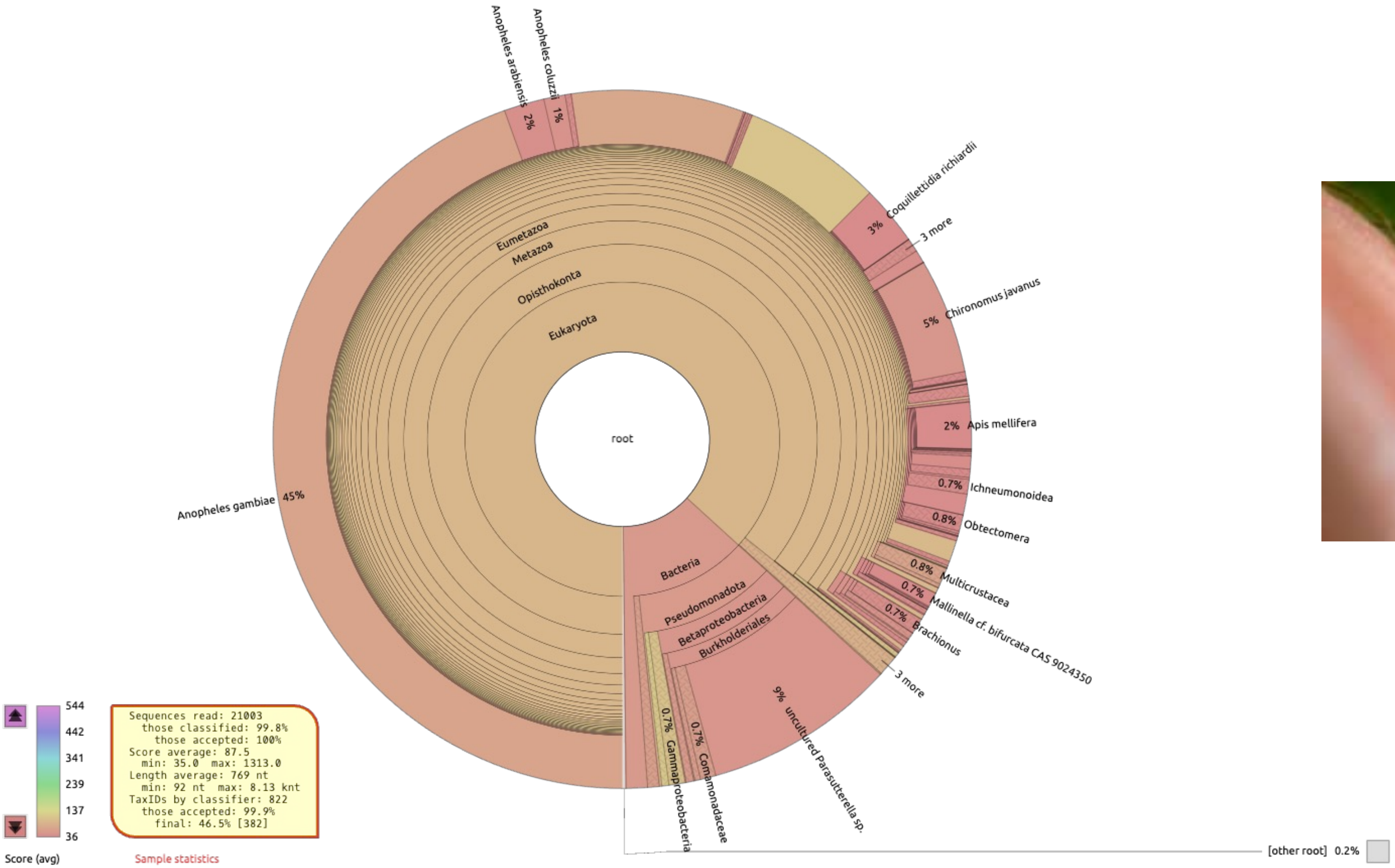
- Unreliable read ends can lead to over-calling of indels
  - *Quality scores across bases / sequences can be calculated (FastQC)*

- Reads can be trimmed / clipped
  - *Hard clipping* does not store the clipped sequence of the read, and is performed when the read is first processed;
  - *Soft clipping* keeps the full read, usually performed within an alignment algorithm

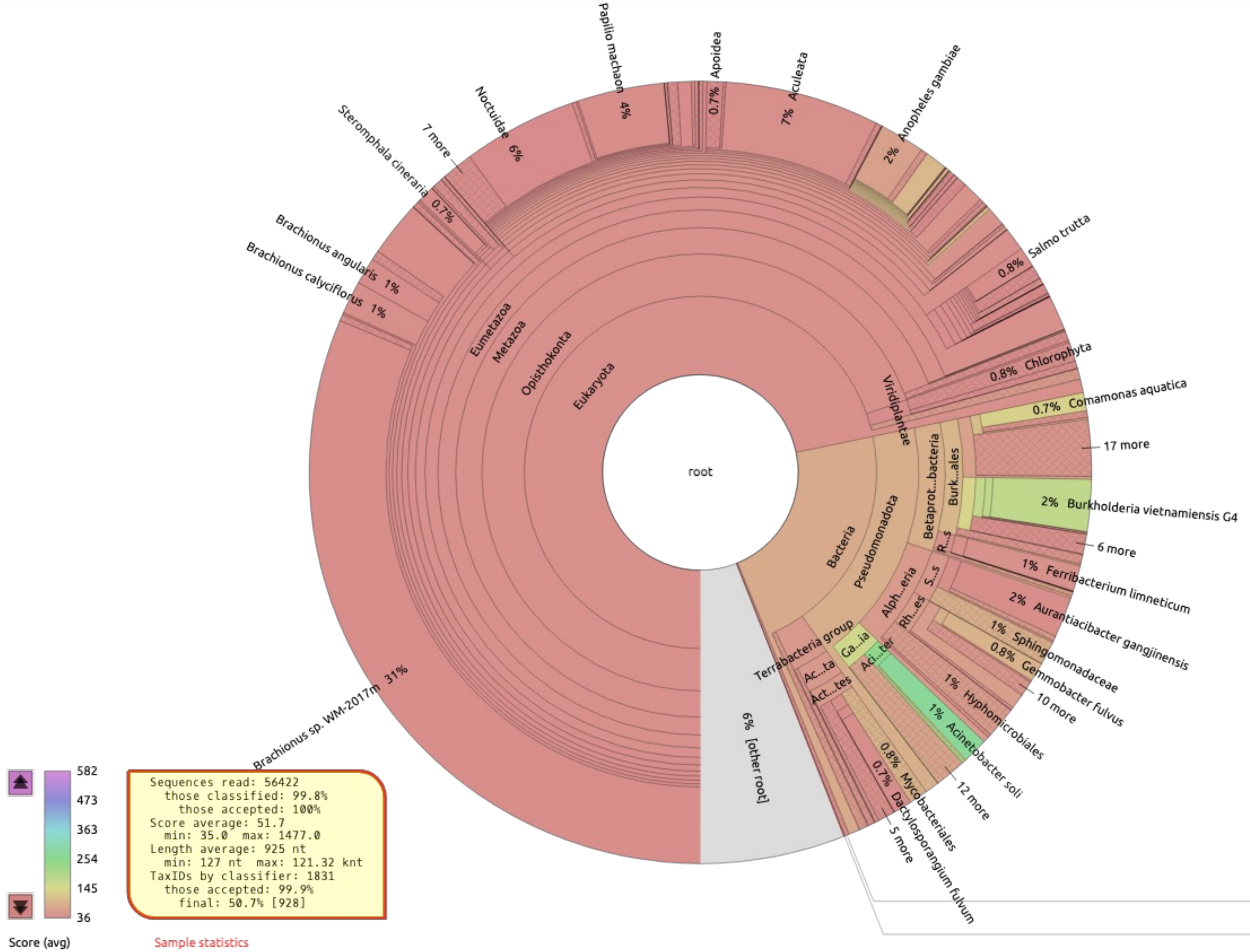
- Duplicate or problematic reads can be filtered (*Picard tools*)

- Mapping statistics
  - % reads mapped: single and paired (*samtools*)
  - % genome that is covered, and to what level (*mosdepth*, *sambamba*)

Example Kraken report



Example Kraken report: Contamination!



37% Brachionus

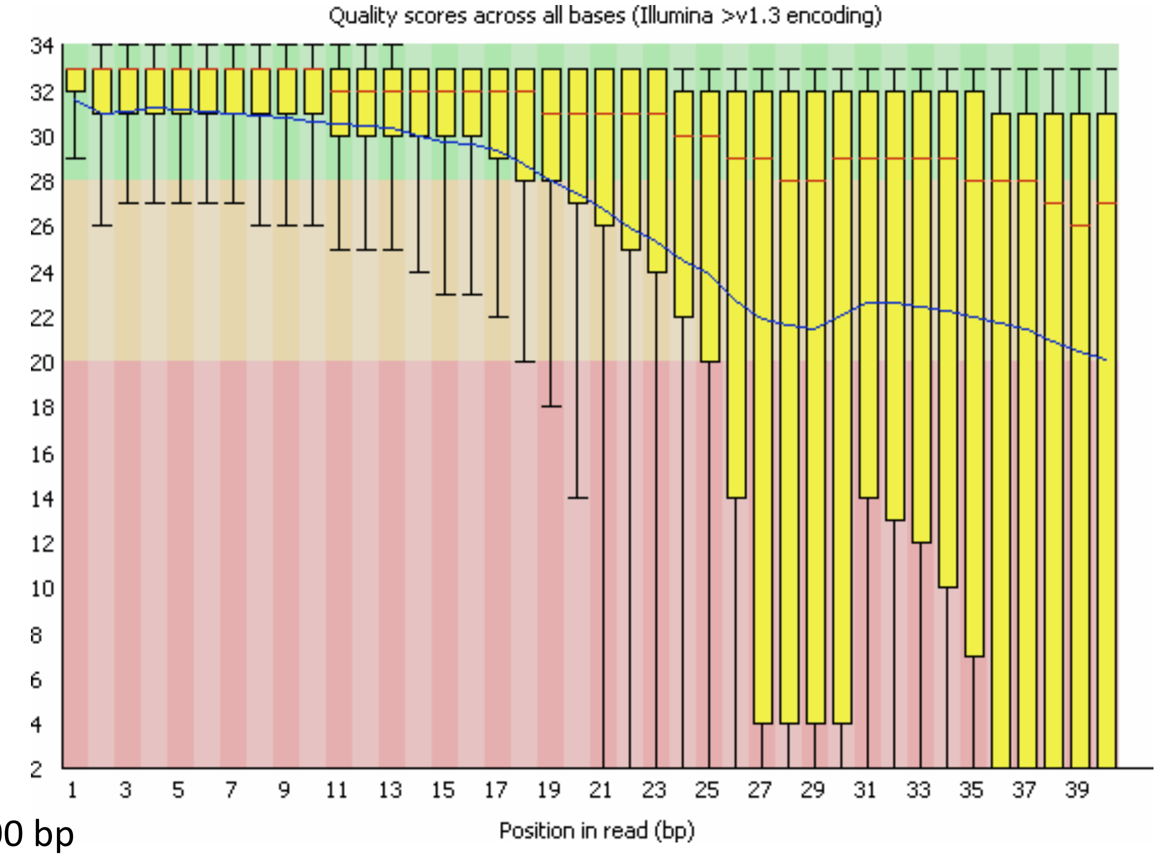
Genus of planktonic rotifers

FastQC: Quality scores are given for every read base

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

Good quality

Poor quality



For good quality you want a score of at least ~20.

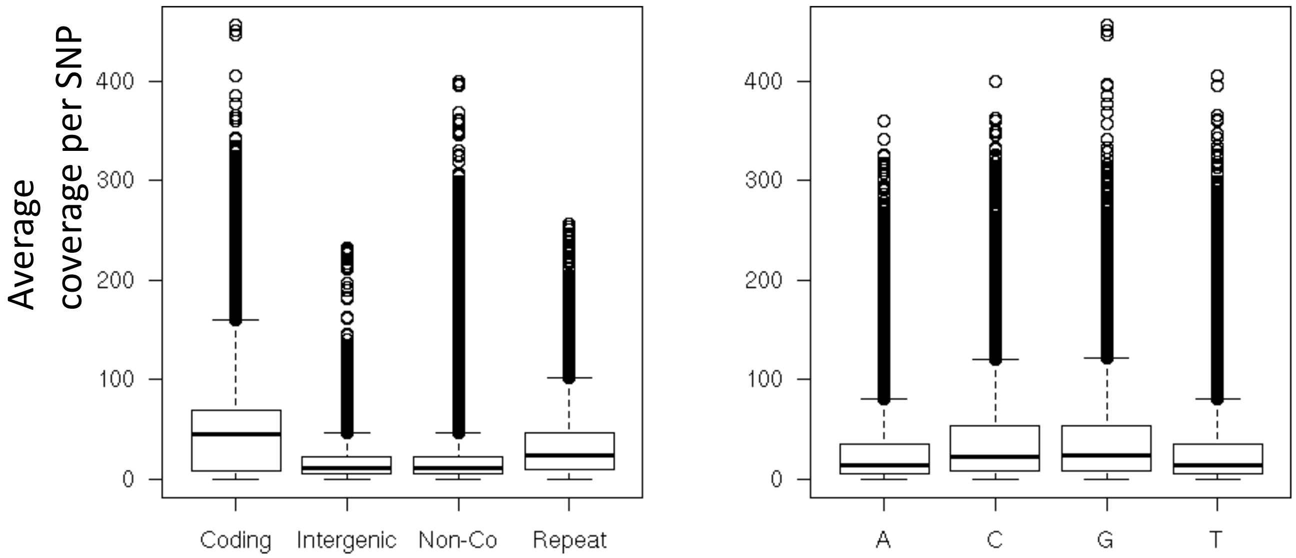
So here, potentially trim reads from 26bp onwards – "trimmomatic" – hard clipping

Q30/40 = 1 error per 1000/10000 bp

Source: Babraham Institute. More information: https://www.youtube.com/watch?v=bz93ReOv87Y

Different factors contribute to poor mapping

1. DNA quality and sequencing errors
2. Inappropriate reference genome
3. Non-unique regions (repeat regions)
4. GC content (for example the *Plasmodium falciparum* genome is 85% AT)



After mapping, the results are stored in specific formats

1. SAM (Sequence Alignment/Map) format.

- See http://genome.sph.umich.edu/wiki/SAM for a description
- It is a TAB-delimited text format consisting of an optional header section, and an alignment section.
- Minimum format agreed on to report sequencing results, and includes all the data in a *fastq* file

2. BAM files (Binary version of SAM)

- Binary version of the SAM format, and much more compact in term of storage

3. CRAM files

- A highly compressed version of a BAM file

Samtools software is used for processing and analysing

- Main software to process and analyse alignments

  (Li et al., 2009, samtools.sourceforge.net)
- Processes BAM (not SAM) files
- Key commands:
  - *Samtools index*: to index the BAM file for rapid analysis
  - *Samtools view*: to extract the genomic region of interest
  - *Samtools flagstat:* to summarise statistics of the mapping
  - *Samtools mpileup:* generates multiple sample alignments in BCF format for variant detection using bcftools

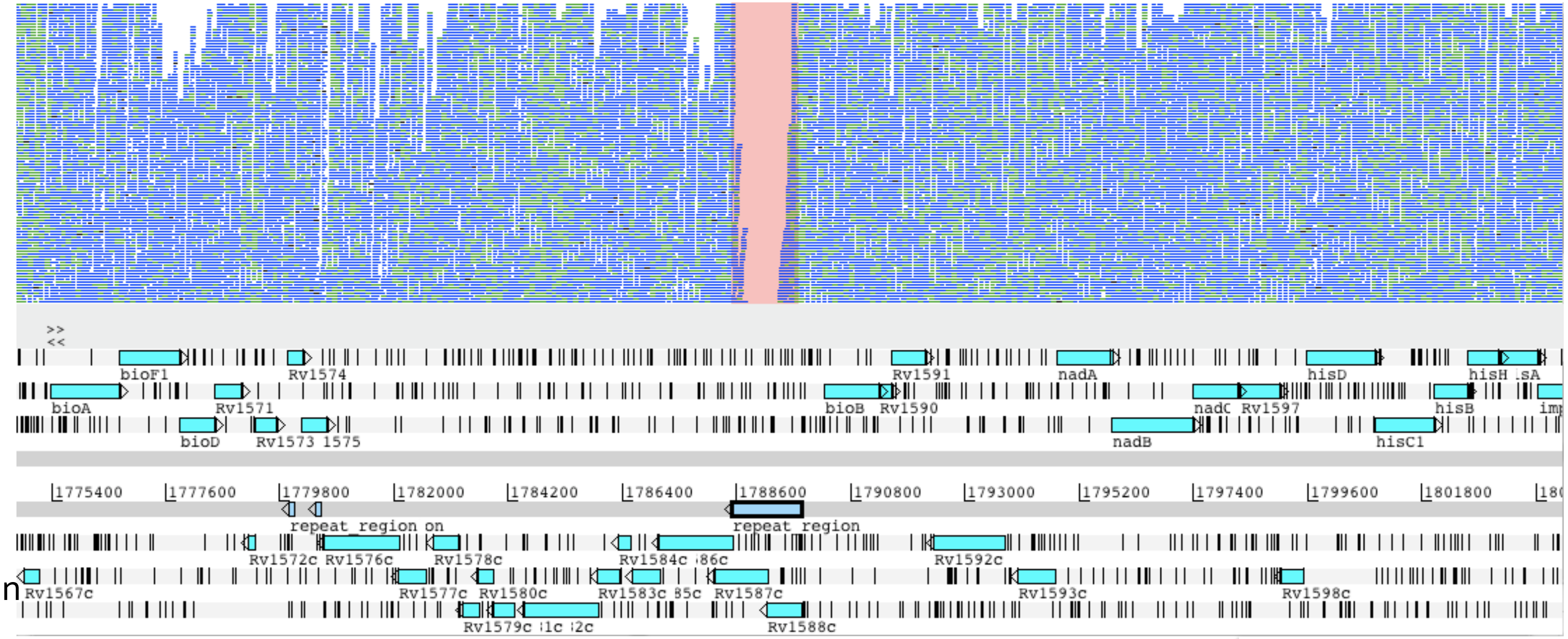
Aligning reads in *Mycobacterium tuberculosis (M. tb)*

LONDON SCHOOL *of* HYGIENE &TROPICAL MEDICINE

Mapping reads to a modified reference (H37Rv) which is GC rich ~65.6%

~97% of the reads mapped uniquely, giving ~300-500 x coverage of the genome

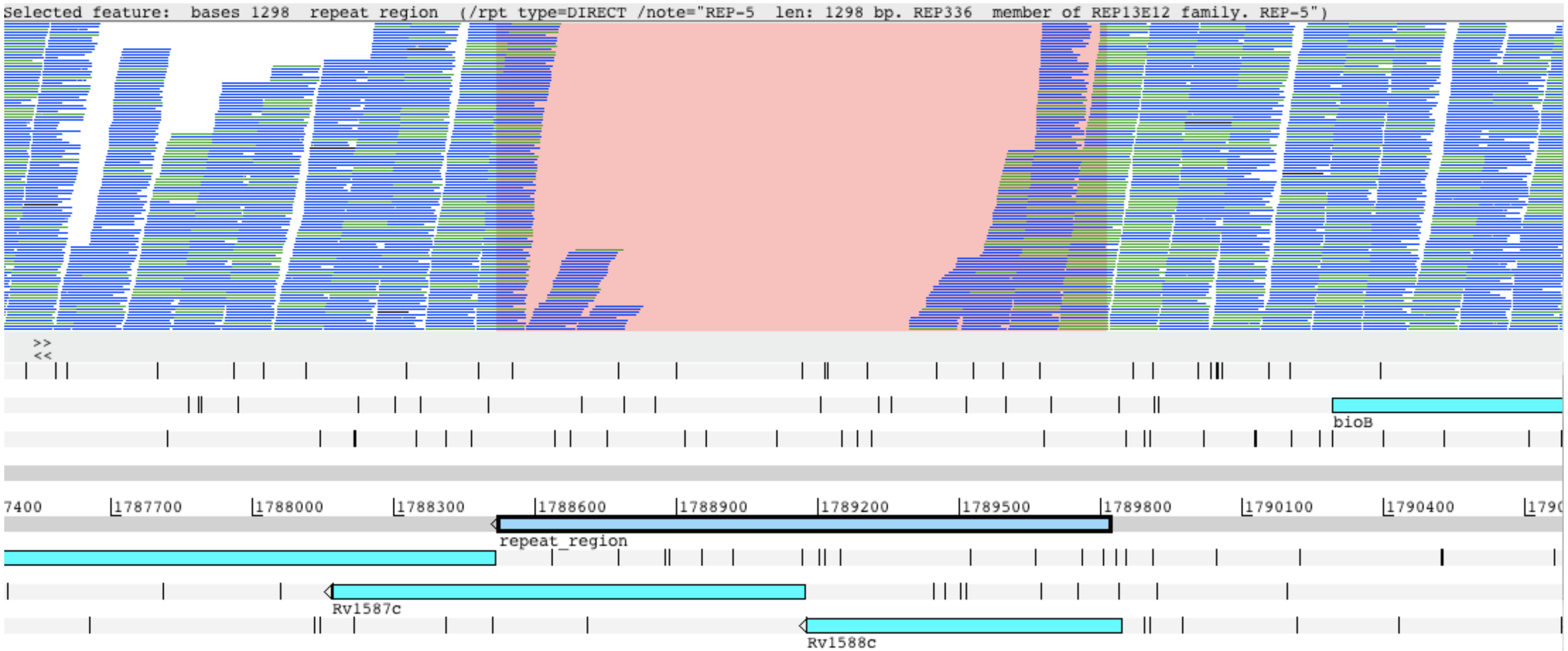
There are some non-unique regions (repeat regions) which are problematic



Artemis software

Lisboa strain of *M. tb*

What if we cannot map to a reference genome?



Solutions:
- Perform *de novo* assembly on non-mapped reads
- Use technologies with longer reads (e.g. ONT Nanopore or PacBio)
- Population-specific reference genomes

Browsing, visualising and interpreting data

- Visualising genome assemblies
  - EagleView [1]
  - HawkEye [2]
  - **Tablet** [3]
- Visualising read alignments with genome annotation
  - LookSeq[4]
  - Integrative Genome Viewer (IGV) **[5]**
  - Integrated Genome Browser (IGB) [6]
  - BamView (integrated with **Artemis**, ACT) [9-11]
  - GenoViewer [12]
- Browsers also with genetic variation detection and analysis
  - MagicViewer [7]
  - Savant [8]

References

1. Huang W, Marth GT. EagleView: a genome assembly viewer for next-generation sequencing technologies. Genome Res 2008;18:1538–43.
2. Schatz MC, Phillippy AM, Shneiderman B, Salzberg SL. Hawkeye: an interactive visual analytics tool for genome assemblies. Genome Biol 2007;8:R34.
3. Milne I, Bayer M, Cardle L, et al. Tablet—next generation sequence assembly visualization. Bioinformatics 2010;26: 401–2.
4. Manske HM, Kwiatkowski DP. LookSeq: a browser-based viewer for deep sequencing data. Genome Res 2009;192125–32.
5. Robinson JT, Thorvaldsdottir H, Winckler W, et al. Integrative genomics viewer. Nat Biotech 2011;29:24–6.
6. Nicol JW, Helt GA, Blanchard SG, et al. The Integrated Genome Browser: free software for distribution and exploration of genome-scale datasets. Bioinformatics 2009;25:2730–1.
7. Hou H, Zhao F, Zhou L, et al. MagicViewer: integrated solution for next-generation sequencing data visualization and genetic variation detection and annotation. Nucleic Acids Res 2010;38:W732–6.
8. Fiume M, Williams V, Brook A, et al. Savant: genome browser for high-throughput sequencing data. Bioinformatics 2010;26:1938–44.
9. Carver T, Bo¨hme U, Otto TD, et al. BamView: viewing mapped read alignment data in the context of the reference sequence. Bioinformatics 2010;26:676–7.
10. Rutherford K, Parkhill J, Crook J, et al. Artemis: sequence visualization and annotation. Bioinformatics 2000; 16:944–5.
11. Carver T, Berriman M, Tivey A, et al. Artemis and ACT: viewing, annotating and comparing sequences stored in a relational database. Bioinformatics 2008;24:2672–6.
12. http://www.genoviewer.com

Mapping Practical

P. falciparum and *M. tuberculosis*

LONDON SCHOOL *of* HYGIENE &TROPICAL MEDICINE